

External wrench → CRG allocation → WBC tracking — tmp

$\rho=1$; admittance scale=1; $K_{a,xyz}=200$ N/m; pre-allocation $|\delta_{xc,xyz}| \leq 50$ mm; steady per-axis $F_{\text{limit}}=10$ N; nonzero-wrench-window QP/Ab valid = 100%/100%

